



Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491 Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K.Jeyakarthika, AP/CSE

Date of Implementation : 31.08.2023

Innovative Practice Description

- **Unit / Topic:** Unit II / Advanced Encryption Standard
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Think-Pair-Share
- **Justification:**
 - Advanced Encryption Standard Implementation Process and detailed structure is such an interesting and important topic, the students are eager to discuss and share their ideas. This practice assists students in gaining a thorough understanding of this topic and real-time application projects. Students think about a topic on their own and express their thoughts with their classmates, which helps them enhance their creative thinking and oral communication abilities.

Time Allotted for the Activity: 20 minutes

Details of the Implementation

- Think-Pair-Share innovative practice conducted for III year CSE students, after explanation of the topic AES for 16- byte key.
- First, I asked the students to think about the working of AES, and how it works for 24- byte key and 32- byte key
- The students referred the materials, discussed and compare about what is the difference between implementation of AES with 16 – byte key, 24-byte key and 32- byte key.
- Then I make them as a pair and share their thinking with their partners.
- Finally, I asked any two or three pairs to explain the concept to whole class for further discussion.
- The Figure 1 and 2 shows the Think Pair Share activity done by the students.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO12
CO1	3	2	1	2	2	-	-	1	1	1

(1 – Low 2 – Moderate 3 – High)



PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12
	3	2	1	2	2	1	1	1
	Basic Computer Science Engineering and mathematics Knowledge for performing Symmetric Encryption is highly correlated	Analyzing various Symmetric encryption algorithms is highly correlated	Developing systems with basic Symmetric encryption mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated.	Use of virtual environment and crypt tool for performing symmetric encryption is highly correlated	Students will complete the assignments/tasks in an ethical way	The students will work effectively as an individual and team for solving the problems in Symmetric Encryption Algorithms.	Updating the Symmetric Encryption algorithms to the current trends are moderately correlated

- Images / Screenshot of the practice:

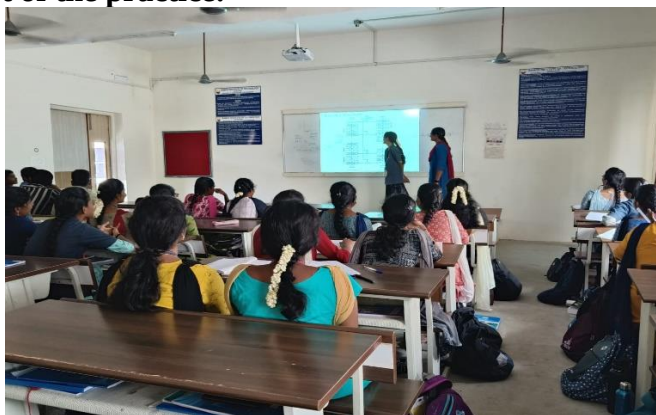


Figure 1 – Students Think and Share their Thoughts and Topic Presentation

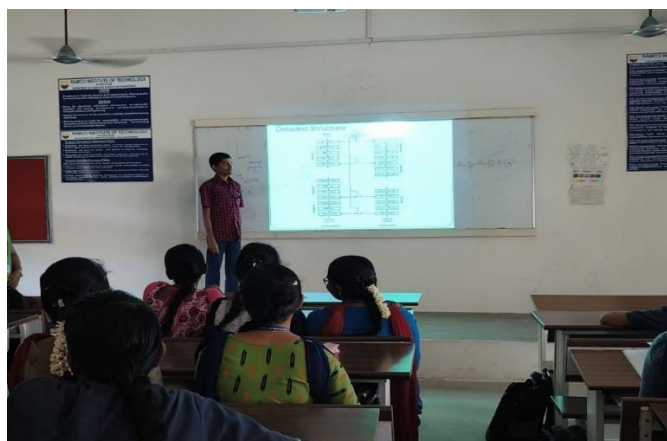




Figure 2 – Students Think and Share their Thoughts and Topic Presentation

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.
 - Different types of reasoning/justification based questions are discussed in the class room, so students come to know how to answer/draw diagrams whenever new real time system is asked in the question

- ❖ ***Benefit of the practice:***

- Students are actively engaged.
 - Students learn from each other.
 - Builds a friendly, yet academic atmosphere.
 - Includes all the students in the teaching-learning process.

- ❖ ***Challenges faced in implementation:***

- Students felt very difficult in this activity and could not complete in time whenever new real time system asked in the question.
 - Students are hesitating to speak in front of all.
 - It took a longer time than other strategies.

References:

- ❖ <http://www.et.iitb.ac.in/TeachingResources.html>
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>
- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improveyour-classroom-discussions1704.html> ∅ <https://serc.carleton.edu/introgeo/interactive/tpshare.html>

Signature of Faculty Member

HOD-CSE